SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Bhavatharani. H (192126025 6) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Bhavi.

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	1	2
Application & Analysis	20%	1	1	2
Explanation & Clarity	15%	1	2	1

Total Marks 25 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

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25/30

CO-1 - AT-2

CONCEPT - MAPPING

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CO1 - AT - 2 - CSA07

CONCEPT - MAPPING

1)

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1. Missing layers

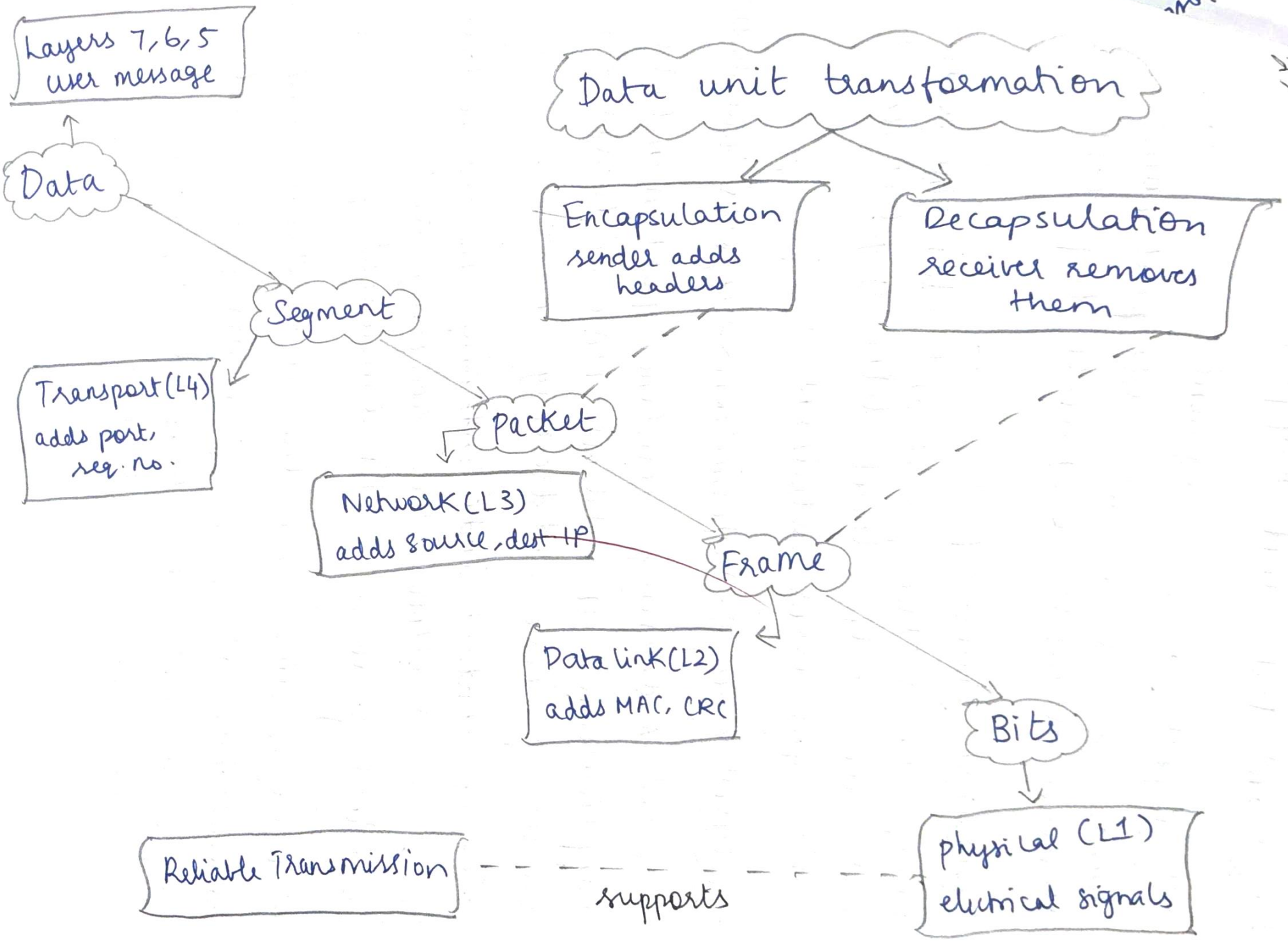
→ Session (L5) - sets up, manages and ends sessions. It gives dialog control and synchronisation check-points, so a broken transfer resumes instead of restarting.

→ Data Link (L2) - packs bits into frames with MAC addresses. It detects errors (CRC) and controls flow and access.

→ Importance - session keeps communication organised and recoverable, and Data link makes each hop error-free.

Table- Missing Layers

Layer	Function	Importance
Session (L5)	Manages sessions, dialog control, Checkpoints	Transfer resumes after failure
Data Link (L2)	Frames, MAC address, CRC error detection	Error-free hop-to-hop link


Layers 7,6,5
user message
Data unit transformation
Data
Encapsulation
sender adds
headers
Decapsulation
receiver removes
them
Segment
Transport (L4)
adds port,
seq. no.
Packet
Network (L3)
adds source, dest IP
Frame
Data link (L2)
adds MAC, CRC
Bits
Reliable Transmission
supports
physical (L1)
electrical signals

Layer and Data Unit Mapping

→ Transport Layer → Segment :
chunks raw data and adds port numbers and sequence numbers for process to process delivery

→ Network layer → Packet :
Encapsulates the packet and adds MAC addresses for node-node local delivery plus an FCS trailer for error detection.

Analysis - How Transformation supports Transmission

→ Tiered Addressing - Combines port numbers (application), IP address, MAC addresses to ensure precise addressing.

→ Error control and Integrity - The Frame trailer (CRC/FCS) allows the receiving node to discard corrupted bits immediately.

→ Decapsulation - At the destination, each layer strips its corresponding header progressively upward until the original application message is successfully reconstructed.


CROSS-LINK-TWO-CLUSTERS
Data Link Layer 2
Network Layer 3
uses
uses
MAC address changes at every hop
works with
IP address Logical, never changes
ensures
ensures
Local Delivery same network only
Global delivery Across networks
ensures
ensures
Complete data delivery Router rewrites MAC per hop
IP addresses PC to server, unchanged
PC
MAC1
Router 1
MAC2
Router 2
MAC3
Server

Core concepts:

- Network Layer → IP Address → Global delivery

Operates at Layer 3 using globally unique logical IP address. Routers inspect destination IPs to route packets across multiple interconnected networks from source to final destination.

- Data Link Layer → MAC Address → Local Delivery

Operates at layer 2 using physical MAC address embedded in network interface cards.

Switches use MAC address to forward frames locally between adjacent nodes on the same physical link.

1. Source setup - Host A creates a packet with constant source and Destination IP address

2. First Hop Framing - Host A wraps the packet in a frame using its own source MAC and the default router's local MAC address.

3. Hop-by-Hop Routing - As the packet moves across routers, IP address remains constant, but MAC addresses are rewritten at every router hop to across local links.